

CS 3700 — Lab 1: Conceptual Data Modeling with ER Diagrams

Due: 11:59 PM Sep 18, 2026

Submit to: Pilot Dropbox

The goal of this lab is to practice **conceptual database design** using the **Entity–Relationship (ER) model**.

You will design an **ER diagram** that models a real-world system **before** any tables or SQL are created.

- This lab focuses on **thinking like a database designer**, not a programmer.

Learning Objectives

By completing this lab, you will be able to:

- Identify **entities, attributes, and relationships** from a problem description
- Select appropriate **primary keys**
- Apply **cardinality** and **participation constraints**
- Draw a correct **ER diagram using Chen notation**
- Distinguish **conceptual design** from implementation details

Business Scenario: Clinic Management System

A medical clinic wants to build a **database system** to manage its daily operations.

The clinic currently relies on spreadsheets and paper forms, which have led to:

- Duplicate records
- Scheduling conflicts
- Inconsistent information

Before building any software, the clinic wants a **proper conceptual database design**.

Your task is to design the **ER model** for this system.

System Requirements

The clinic needs to keep track of:

- **Patients**
 - **Doctors**
 - **Departments**
 - **Appointments**
-

Patient

For each **Patient**, store:

- PatientID
- Name
- DateOfBirth
- Phone
- Email

Each patient is uniquely identified by **PatientID**.

Doctor

For each **Doctor**, store:

- DoctorID
- Name
- Specialty
- OfficeNumber

Each doctor is uniquely identified by **DoctorID**.

Department

For each **Department**, store:

- DepartmentID
- DepartmentName
- Location

Each department is uniquely identified by **DepartmentID**.

Appointment

Appointments represent **visits between patients and doctors**.

For each **Appointment**, store:

- AppointmentID
- Date
- Time
- Reason

Each appointment is uniquely identified by **AppointmentID**.

Business Rules

1. Each **Doctor** works in **exactly one Department**.
 2. Each **Department** can have **many Doctors**.
 3. A **Patient** can have **many Appointments**.
 4. A **Doctor** can have **many Appointments**.
 5. Each **Appointment** is with **exactly one Patient** and **exactly one Doctor**.
 6. A **Patient** may exist **without having any appointments yet**.
 7. A **Doctor** may exist **without having any appointments yet**.
 8. An **Appointment cannot exist** without both a Patient and a Doctor.
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Your Tasks

Part 1 — Identify Components (20 points)

List the following:

- All **entities**
 - All **attributes** for each entity
 - The **primary key** for each entity
 - All **relationships** between entities
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Part 2 — Determine Constraints (20 points)

For **each relationship**, clearly specify:

- **Cardinality** (1:1, 1:N, M:N)
- **Participation** (Total or Partial on each side)

Write your answers in words.

Example:

Each Appointment must be associated with exactly one Patient (total participation).
A Patient may have zero or more Appointments (partial participation).

Part 3 — Draw the ER Diagram (50 points)

Draw a **complete ER diagram** using **Chen notation** that includes:

- All entities
- All attributes
- Clearly marked **primary keys**
- All relationships
- Cardinality constraints
- Participation constraints

You may use:

- diagrams.net (draw.io)
- Lucidchart
- PowerPoint / Google Slides
- Any ER diagram tool

You **may not** submit a hand-drawn image.

Part 4 — Design Justification (10 points)

Answer **both** questions:

1. Why is **Appointment** modeled as an **entity** instead of just a relationship?
2. What would be lost or break if **Appointment** were not an entity?

Write **2–5 sentences per question**.

Submission Instructions

Submit **one PDF file** that contains:

1. Entity & relationship list
 2. Constraint explanations
 3. ER diagram
 4. Written justification answers
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Grading Rubric (100 Points)

Category	Description	Points
Entities & Attributes	Correct identification of entities and attributes	20
Keys	Correct primary key selection and marking	10
Relationships	All required relationships are correctly modeled	10
Cardinality Constraints	Correct 1:1, 1:N, M:N constraints	15
Participation Constraints	Correct total vs partial participation	15
ER Diagram Quality	Clear, readable, correct Chen notation	20
Design Justification	Clear and correct written explanations	10

Category	Description	Points
Total		100
